

LOWER CARP LAKE, NT, Canada

WMO: 716800

Lat: 63.600N

Lon: 113.850W

Elev: 1225

StdP: 14.06

Time Zone: -7.00 (NAM)

Period: 98-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.0	(c) -37.0	(d) -48.4	(e) 0.3	(f) -39.9	(g) -44.9	(h) 0.4	(i) -36.8	(j) 21.5	(k) 3.0	(l) 19.6	(m) -3.0	(n) 3.4	(o) 60	(p) 0.828

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	78.2	59.7	74.9	58.1	71.5	56.8	61.6	73.5	60.0	71.3	58.4	68.9	9.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.3	73.2	64.6	55.4	68.4	63.2	53.6	63.8	62.0	28.1	73.8	27.0	71.2	25.9	69.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.6	17.4	-43.4	82.2	2.8	1.8	-45.4	83.5	-47.0	84.5	-48.6	85.6	-50.6	86.9
			-43.6	63.6	2.8	1.6	-45.6	64.8	-47.2	65.7	-48.8	66.6	-50.8	67.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	22.2	-13.9	-10.9	-1.3	17.0	36.5	53.5	61.4	55.2	43.5	26.6	6.2	-10.0	
	DBStd	29.04	14.29	12.42	15.71	13.71	11.33	8.46	6.55	7.01	8.24	8.78	11.55	14.26	
	HDD50	10919	1981	1706	1591	990	435	59	4	29	224	726	1313	1861	
	HDD65	15688	2446	2126	2056	1440	882	354	147	313	645	1190	1763	2326	
	CDD50	757	0	0	0	0	18	164	357	189	29	0	0	0	
	CDD65	52	0	0	0	0	0	9	35	8	0	0	0	0	
	CDH74	368	0	0	0	0	3	80	232	53	0	0	0	0	
	CDH80	47	0	0	0	0	0	13	29	5	0	0	0	0	
Wind	WSAvg	8.1	5.7	6.8	8.0	8.8	9.4	9.2	8.8	9.1	9.8	8.9	6.7	5.5	
Precipitation	PrecAvg	12.2	0.5	0.4	0.6	0.4	0.7	1.4	1.9	2.1	1.7	1.1	0.8	0.6	
	PrecMax	15.9	0.9	0.9	1.3	0.9	1.4	3.8	3.9	4.2	3.7	1.7	1.4	1.1	
	PrecMin	8.6	0.2	0.1	0.1	0.1	0.1	0.2	0.7	0.7	0.6	0.4	0.4	0.3	
	PrecStd	1.6	0.2	0.2	0.3	0.2	0.4	0.7	0.7	0.7	0.7	0.4	0.2	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	21.2	38.7	52.3	73.0	81.8	83.5	79.3	68.2	54.2	31.7	24.0	
		MCWB	26.0	20.6	33.7	41.6	52.2	59.9	61.4	60.7	55.5	47.3	31.1	23.7	
	2%	DB	21.1	16.4	32.6	45.7	64.0	75.6	79.6	74.8	62.6	44.7	29.1	19.7	
		MCWB	20.6	15.7	28.7	37.8	49.1	56.5	60.8	59.0	52.2	40.9	28.6	19.1	
	5%	DB	14.0	12.8	27.4	41.6	58.5	71.7	76.4	70.8	58.7	40.4	26.2	15.0	
		MCWB	13.3	12.2	24.7	35.2	45.6	55.3	59.5	57.9	50.5	38.0	25.7	14.5	
	10%	DB	6.7	7.9	22.2	37.0	53.6	67.7	73.3	66.9	54.8	36.9	22.3	10.7	
		MCWB	6.1	7.2	20.3	32.2	43.0	53.6	58.0	56.0	48.8	35.2	21.7	10.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.7	20.5	34.1	42.1	53.1	60.8	64.3	62.8	56.6	48.6	31.0	23.8
			MCDB	25.9	21.0	38.1	50.6	68.3	74.5	76.9	74.7	64.9	53.5	31.9	23.9
2%		WB	20.4	15.7	28.7	38.2	49.8	58.6	62.5	60.7	53.6	41.3	28.6	19.1	
		MCDB	20.8	16.3	32.4	45.1	64.2	72.6	74.3	71.7	60.4	44.1	29.0	19.6	
5%		WB	13.4	12.2	24.7	35.5	46.2	56.6	61.0	58.7	51.3	37.9	25.7	14.5	
		MCDB	13.9	12.8	27.3	41.3	57.3	68.6	72.4	68.4	57.6	40.0	26.1	14.9	
10%		WB	6.2	7.3	20.4	32.6	43.3	54.6	59.6	56.9	49.4	35.4	21.8	10.1	
		MCDB	6.7	7.9	22.4	36.6	52.5	66.2	70.6	65.0	54.0	36.8	22.3	10.7	
Mean Daily Temperature Range	5% DB	MDBR	13.1	15.5	19.4	21.4	18.5	17.2	15.8	13.8	11.1	8.5	11.0	11.9	
		MCDBR	13.2	15.8	20.8	22.4	24.8	21.8	20.0	19.9	17.5	12.6	10.3	13.5	
		MCWBR	13.5	15.6	17.2	15.4	13.2	8.5	7.4	8.1	9.1	8.8	10.1	13.5	
	5% WB	MCDBR	13.7	15.5	19.9	21.4	23.3	19.4	17.6	17.8	16.3	12.0	10.2	13.5	
		MCWBR	13.7	15.4	16.6	14.8	12.6	8.5	7.5	8.4	9.1	8.7	10.1	13.6	
Clear-Sky Solar Irradiance	taub	0.174	0.227	0.257	0.288	0.322	0.325	0.353	0.326	0.296	0.262	0.171	0.147		
	taud	1.982	2.235	2.249	2.235	2.398	2.493	2.433	2.512	2.584	2.392	1.988	1.938		
	Ebn at Noon	149	235	271	284	283	285	271	271	260	218	145	88		
	Edh at Noon	9	17	26	34	32	30	31	26	20	15	8	4		
All-Sky Solar Radiation	RadAvg	75	332	900	1606	1892	1908	1682	1248	727	252	71	28		
	RadStd	5	19	53	78	142	166	87	78	67	30	14	2		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		N/A	N/A	+2.39	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page